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Submission Topic	Assignment 4		

Due date of submission	22/10/2020		
Due time of submission	1:00 PM	Name of tutor (if appropriate)	
Group	05	Session attended (if appropriate)	
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Assignment 4

Q1. (a) Proof: $|\mathbb{R}| = c$

$\therefore$ $\mathbb{R}$ is uncountable

$\because$ $\mathbb{Z}_2 = \{0, 1\}$

$\therefore$ $|\mathbb{Z}_2| = 2$

$\therefore$ $\mathbb{Z}_2$ is countable

$\therefore$ $\mathbb{R} \times \mathbb{Z}_2$ is uncountable

$\therefore$ $|\mathbb{R}| = |\mathbb{R} \times \mathbb{Z}_2|$ $\square$

(b) Proof: $\because$ $\mathbb{R}$ is uncountable, $\mathbb{Z}$ is countable

$\therefore$ $\mathbb{R} \times \mathbb{Z}$ is uncountable

$\therefore$ $|\mathbb{R}| = |\mathbb{R} \times \mathbb{Z}|$ $\square$

Q2. (a) Proof: Yes

(1) Closure Suppose that $x, y \in I = [0, 1)$

Then $\varphi(x+y) = x+y-\lfloor x+y \rfloor \in I$

$\because$ $0 \le x+y < 2$

$\therefore$ $0 \le x+y-\lfloor x+y \rfloor < 1$

$\therefore$ $x \oplus y \in I$

Associativity Suppose that $x, y, z \in I = [0, 1)$

Then $\oplus$ is associative since $(x \oplus y) \oplus z = \varphi(x+y) \oplus z = \varphi(\varphi(x+y)+z) = \varphi(\varphi(x+y-\lfloor x+y \rfloor)+z)$

$x \oplus (y \oplus z) = \varphi(x+(y \oplus z)) = \varphi(x+\varphi(y+z)) = \varphi(x+y-\lfloor x+y \rfloor+z)$

Since $x, y, z \in I = [0, 1)$ $x \oplus (y \oplus z) = \varphi(x+(y \oplus z))$

$= \varphi(x+\varphi(y+z)) = \varphi(x+y+z-\lfloor y+z \rfloor)$

$\therefore$ $\varphi(x+y)+z = x+\varphi(y+z)$

$\therefore$ $\varphi(\varphi(x+y)+z) = \varphi(x+\varphi(y+z))$

$\therefore$ $(x \oplus y) \oplus z = x \oplus (y \oplus z)$

Identity The identity of this group is 0

Since $x \oplus 0 = \varphi(x+0)$

$= x+0-\lfloor x+0 \rfloor$

$= x$

$0 \oplus x = \varphi(0+x)$

$= 0+x-\lfloor 0+x \rfloor$

$= x$

for all $x \in I$

(4) Inverses When $x=0$, the inverse of x is 0.

Since $0 \in I$

and $x \oplus 0 = \varphi(0+0)$

$= \varphi(0)$

$= 0-\lfloor 0 \rfloor$

$= 0-0$

$= 0$

$= 0 \oplus x$

When $0<x<1$, the inverse of x is $1-x$

Since $1-x \in I$ whenever $x \in I$

and $x \oplus (1-x) = \varphi(x+1-x)$

$= \varphi(1)$

$= 1-\lfloor 1 \rfloor$

$= 1-1$

$= 1$

$= (1-x) \oplus x$ □

(b) No.

Proof (by contradiction): It has no identity element in I

~~Since it has an identity element in I,~~

Since $\exists y \in I$ such that $\forall x \in I$,

$y \odot x = x \odot y = x$

$\varphi(yx) = \varphi(xy) = x$

$xy - \lfloor xy \rfloor = x$

$\therefore 0 \le x < 1$

$\therefore y = 1 \notin I$

So we have a contradiction. □

Q3. (a)

$\cdot$	1	i	-1	$-i$
1	1	i	-1	$-i$
i	i	-1	$-i$	1
-1	-1	$-i$	1	i
$-i$	$-i$	1	i	-1

(b) Prove: An isomorphism is given by $f(n) = i^n$, $n \in \mathbb{Z}_4$.

$\therefore \mathbb{Z}_4 = \{0, 1, 2, 3\}$

$\therefore |\mathbb{Z}_4| = 4$

$f(0) = i^0$

$= 1 \in \{1, -1, i, -i\}$

$f(1) = i^1$

$= i \in \{1, -1, i, -i\}$

$f(2) = i^2$

$= -1 \in \{1, -1, i, -i\}$

$f(3) = i^3$

$= i^2 \cdot i$

$= -1 \cdot i$

$= -i \in \{1, -1, i, -i\}$ $\square$

(c) Prove: ~~For a contradiction, suppose~~ $f: \mathbb{C} - \{0\} \to \mathbb{R} - \{0\}$ is an isomorphism

Since $\mathbb{R} - \{0\} = \{1, -1\}$

$\therefore |\mathbb{R} - \{0\}| = 2$

Also: $|\mathbb{Z}_4| = 4$

$\therefore |\mathbb{R} - \{0\}| \neq |\mathbb{Z}_4|$

And this is a contradiction because f is a bijection.

Thus, $(\mathbb{C} - \{0\}, \times)$ is not isomorphic to $(\mathbb{Q} - \{0\}, \times)$. $\square$

Q4. (a) I can vote in 326 possible ways if there are five candidates on the ballot.

The number of ~~number~~ 0 candidate $= 1$

The number of number 1 candidate $= 5$

The number of number 2 candidates $= \frac{5!}{(5-2)!}$

$= 20$

The number of number 3 candidates $= \frac{5!}{(5-3)!}$

$= 60$

The number of number 4 candidates $= \frac{5!}{(5-4)!}$

$= 120$

The number of number 5 candidates $= 5!$

$= 120$

I can vote in $1+5+20+60+120+120=326$ possible ways if there are five candidates on the ballot.

(b) I can vote in 64 possible ways if there are six candidates on ballot.

The number of place 0 candidate $= 1$

The number of place 1 candidate $= 6$

The number of place 2 candidates $= \binom{6}{2}$

$= 15$

The number of place 3 candidates $= \binom{6}{3}$

$= 20$

The number of place 4 candidates $= \binom{6}{4}$

$= 15$

The number of place 5 candidates $= \binom{6}{5}$

$= 6$

The number of place 6 candidates $= 1$

I can vote in $1+6+15+20+15+6+1=64$ possible ways if there are six candidates on ballot.